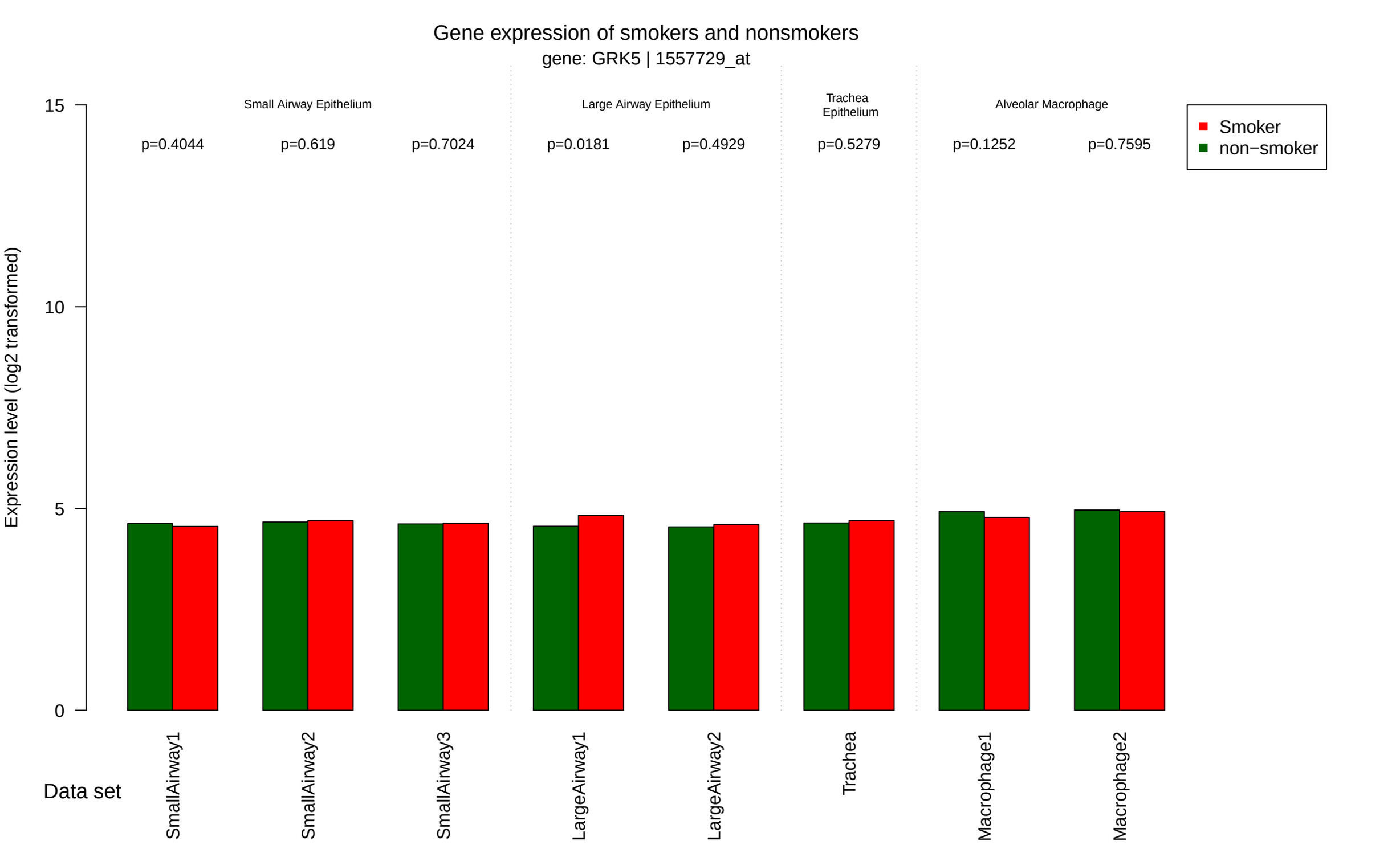
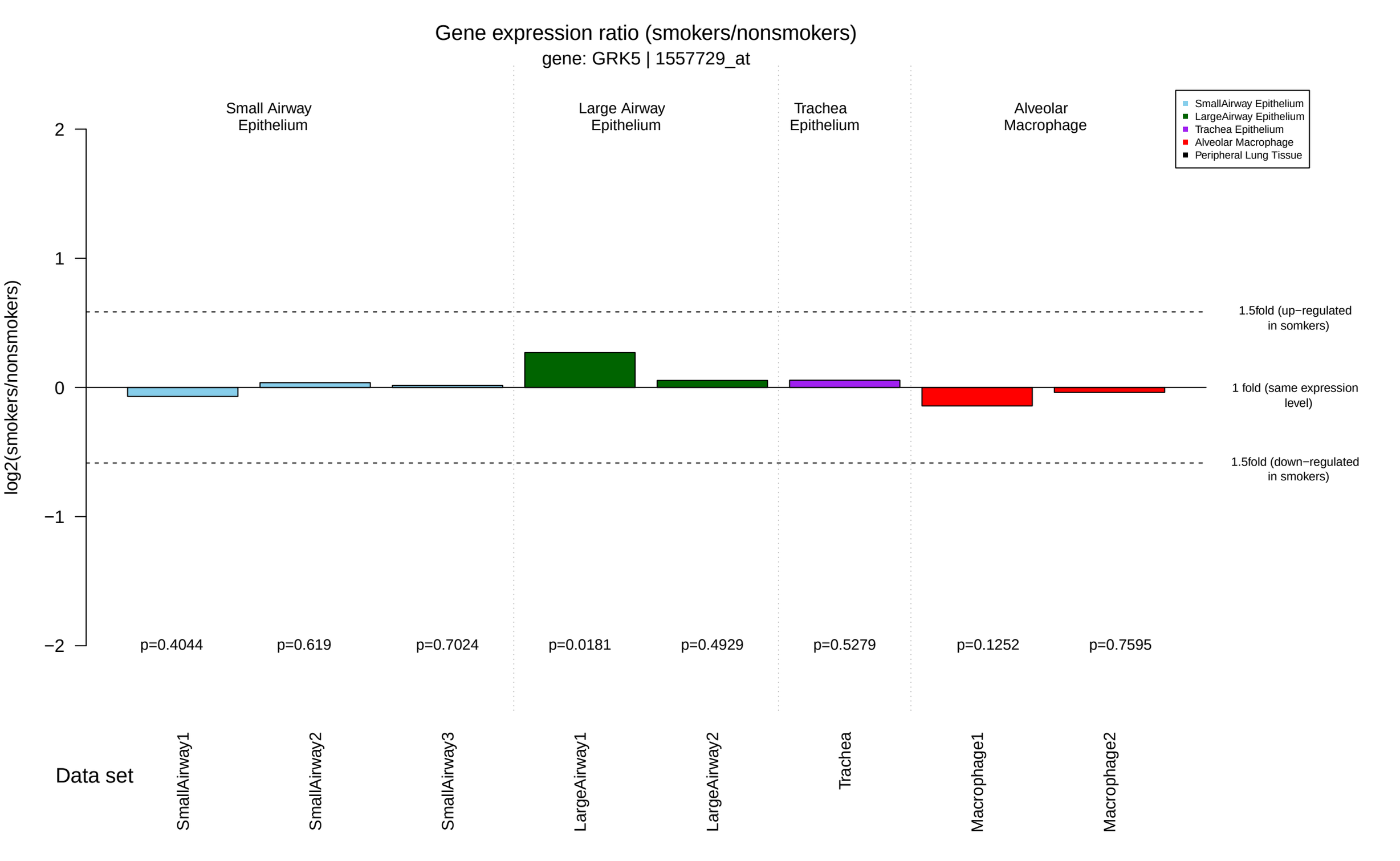
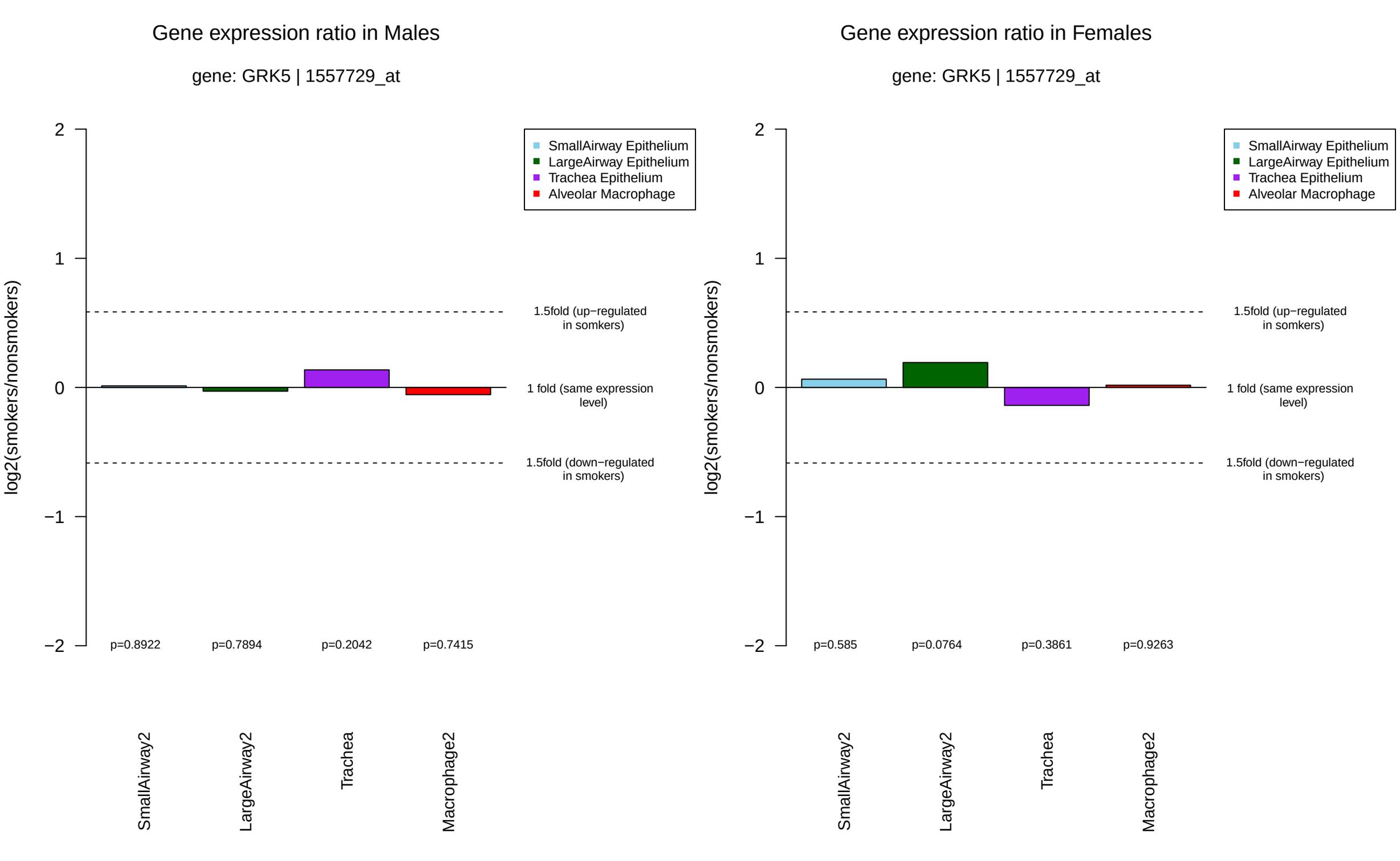
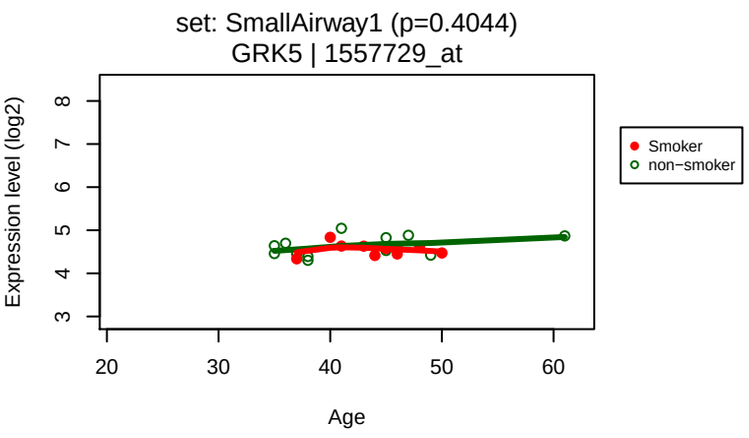
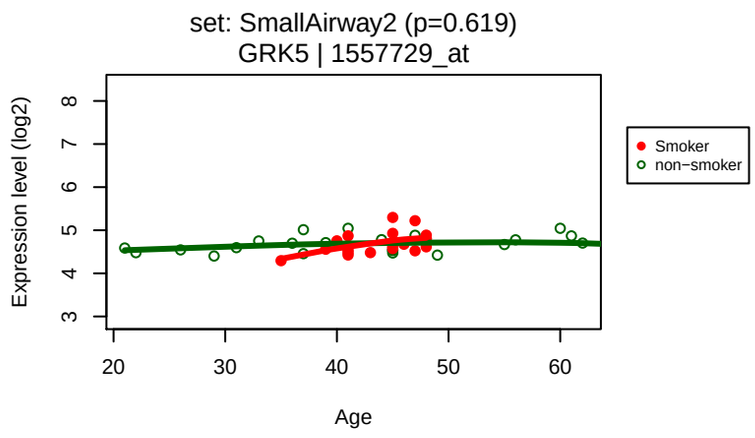
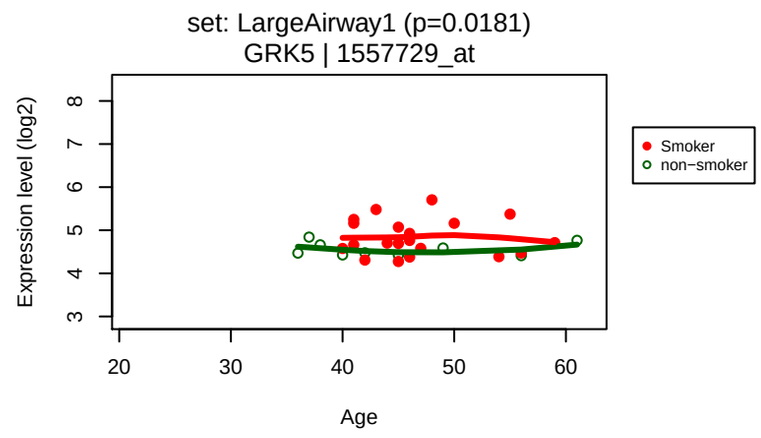

Gene expression of smokers and nonsmokers
gene: GRK5 | 1557729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4044
p=0.619
p=0.7024
p=0.0181
p=0.4929
p=0.5279
p=0.1252
p=0.7595
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

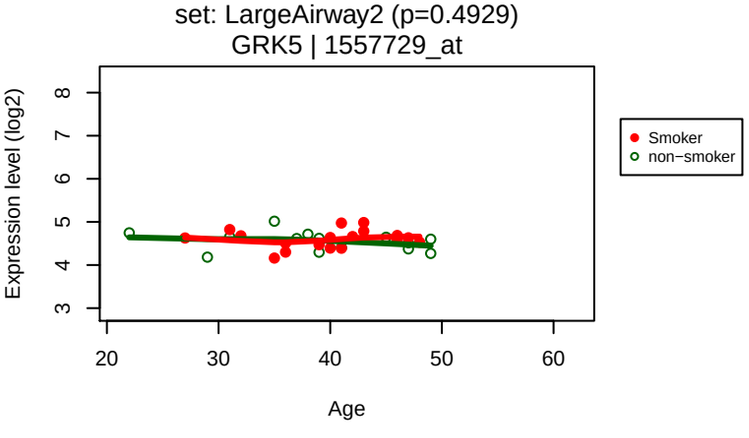

Gene expression ratio (smokers/nonsmokers)
gene: GRK5 | 1557729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4044
p=0.619
p=0.7024
p=0.0181
p=0.4929
p=0.5279
p=0.1252
p=0.7595
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

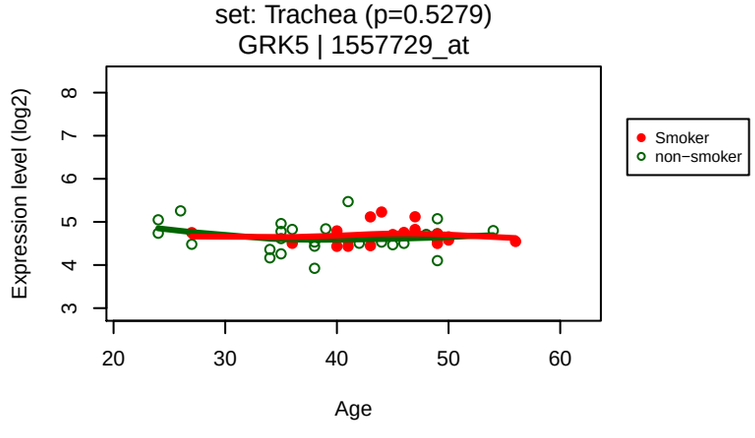

Gene expression ratio in Males
gene: GRK5 | 1557729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8922
p=0.7894
p=0.2042
p=0.7415
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK5 | 1557729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.585
p=0.0764
p=0.3861
p=0.9263
SmallAirway2
LargeAirway2
Trachea
Macrophage2

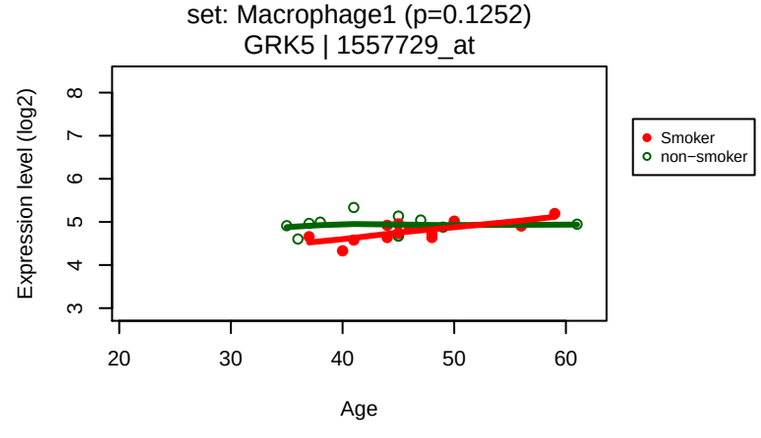

set: SmallAirway1 (p=0.4044)
GRK5 | 1557729_at
Expression level (log2)
Age
Smoker
non−smoker

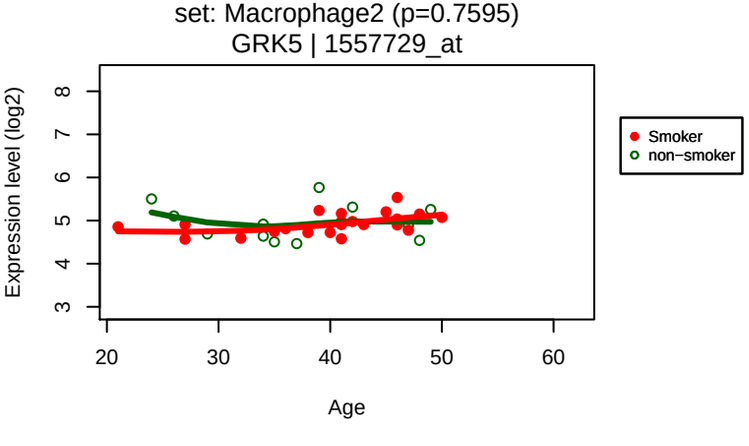

set: SmallAirway2 (p=0.619)
GRK5 | 1557729_at
Expression level (log2)
Age
Smoker
non−smoker

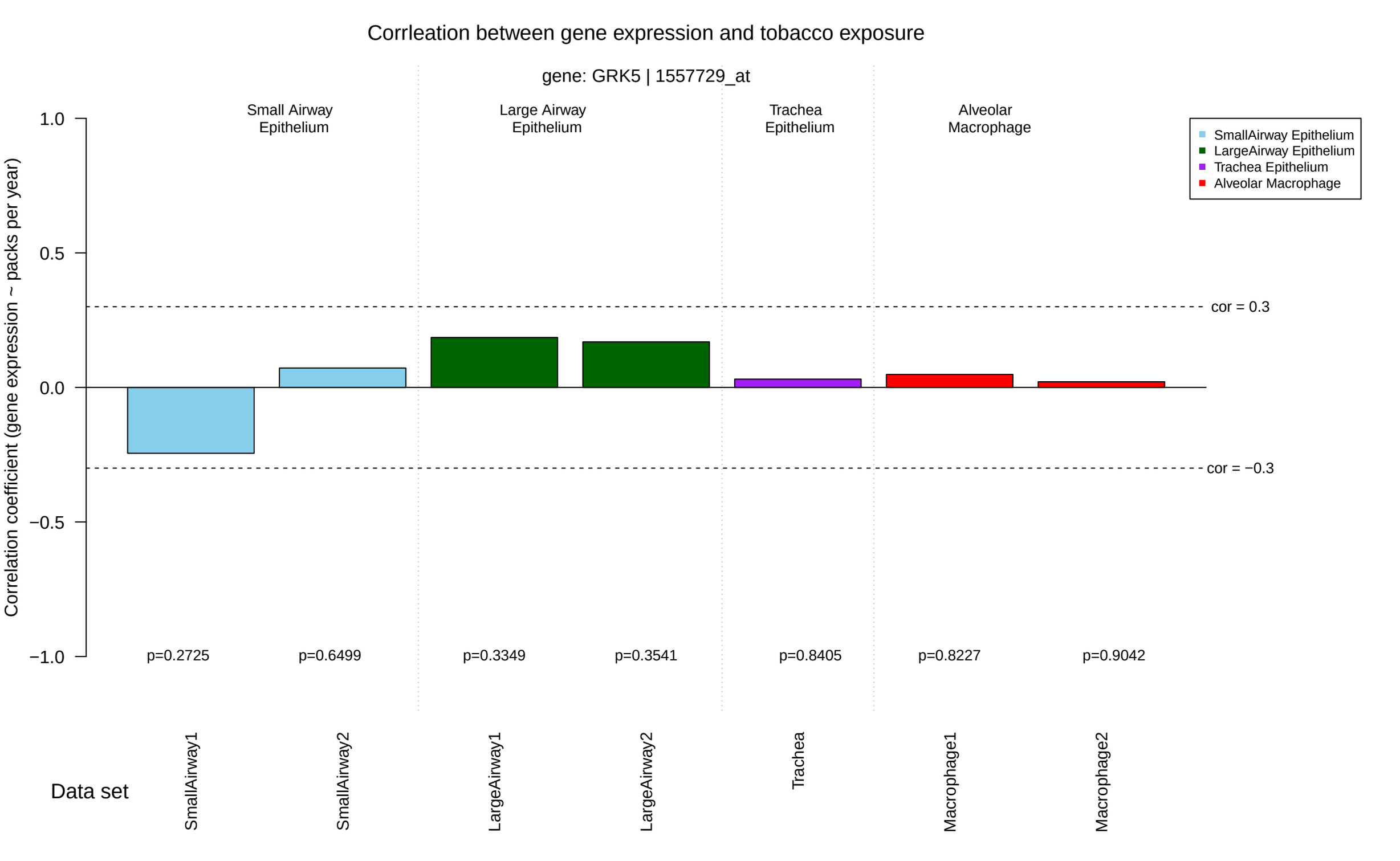

set: LargeAirway1 (p=0.0181)
GRK5 | 1557729_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4929)
GRK5 | 1557729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5279)
GRK5 | 1557729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1252)
GRK5 | 1557729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7595)
GRK5 | 1557729_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK5 | 1557729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2725
p=0.6499
p=0.3349
p=0.3541
p=0.8405
p=0.8227
p=0.9042
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2